

Comparison of Classical, Neural ODE/SDE, and Logistic PINN Fits for Water Kefir Data

Abstract

This report compares five dynamical descriptions of water kefir fermentation data: a classical logistic ordinary differential equation, a deterministic Neural ODE, a Neural SDE, a deterministic logistic PINN, and a stochastic logistic PINN/SDE. The data consist of three trials measured at 15 time points, giving 45 scalar observations. The Neural ODE and the two logistic PINNs are treated as population models: the three trials are interpreted as repeated realizations of the same experiment, so one fitted curve is compared with all trial observations. The stochastic logistic PINN additionally identifies a multiplicative diffusion parameter and generates Monte Carlo predictive intervals. The reported residual criteria are in-sample diagnostics; out-of-sample MSE should be computed with a time-respecting train/test split.

1 Data and preprocessing

Let $y_i^{[j]}$ denote the response observed at time t_i in trial j , with $i = 0, \dots, T-1$ and $j = 1, \dots, J$. In this experiment $T = 15$ and $J = 3$. The time points run from 0 to 168 in the original time units. The observed index set is

$$\Omega = \{(i, j) : y_i^{[j]} \text{ is observed}\}, \quad n = |\Omega| = 45.$$

The Neural ODE and Neural SDE were trained on normalized time and normalized response:

$$\tau_i = \frac{t_i - t_{\min}}{t_{\max} - t_{\min}}, \quad z_i^{[j]} = \frac{y_i^{[j]} - \bar{y}}{s_y},$$

where

$$\bar{y} = \frac{1}{n} \sum_{(i,j) \in \Omega} y_i^{[j]}, \quad s_y = \left[\frac{1}{n} \sum_{(i,j) \in \Omega} (y_i^{[j]} - \bar{y})^2 \right]^{1/2}.$$

Thus $\tau_i \in [0, 1]$. The logistic PINN models also use normalized time, but their neural trajectories are expressed directly on the original response scale. Their data and physics losses are divided by s_y^2 so that residual terms are dimensionless and numerically comparable. All model diagnostics were computed on the original response scale; for the Neural ODE and Neural SDE this required transforming fitted normalized values back to the scale of $y_i^{[j]}$.

2 Models

2.1 Classical logistic ODE

The classical baseline is the Verhulst logistic growth model [10], fitted here as a shared-parameter nonlinear regression model. Unlike the neural models, it is fitted directly to the original response

values $y_i^{[j]}$. The only normalization used by this baseline is the normalized time variable $\tau_i \in [0, 1]$. Each trial has its own observed initial condition, fixed at the first measured value, but all trials share the same growth rate and carrying capacity:

$$\frac{dy^{[j]}}{d\tau} = ry^{[j]} \left(1 - \frac{y^{[j]}}{K} \right), \quad y^{[j]}(0) = y_0^{[j]} = y^{[j]}(t_0).$$

The exact solution used in the fit is

$$y^{[j]}(\tau) = \frac{K}{1 + \left(\frac{K - y_0^{[j]}}{y_0^{[j]}} \right) \exp(-r\tau)}.$$

The unknown parameters of the classical fit are only r and K ; the initial values $y_0^{[j]}$ are taken from the data and are not optimized. The fit also requires positive observations, because the closed-form logistic curve and the ratio $(K - y_0^{[j]})/y_0^{[j]}$ are only meaningful for positive initial values. Missing observations, if present, are excluded through the observed index set Ω .

The fitted parameters must satisfy $r > 0$ and $K > \max_j y_0^{[j]}$. Instead of optimizing constrained parameters directly, the implementation optimizes two unconstrained real variables $a, b \in \mathbb{R}$ and maps them to valid physical parameters with a softplus transformation:

$$r = \text{softplus}(a) + 10^{-6}, \quad K = K_{\min} + \text{softplus}(b) + 10^{-6},$$

where $K_{\min} = \max_j y_0^{[j]} + 10^{-3}$ and

$$\text{softplus}(x) = \log(1 + \exp(x)).$$

The softplus function is a smooth differentiable approximation to the positive part function. It maps every real number to a positive value, behaves approximately like $\exp(x)$ for large negative x , and behaves approximately like x for large positive x . This is useful for gradient-based fitting: Adam can update the unconstrained variables a and b freely, while the reported parameters r and K remain positive and the carrying capacity stays above every initial condition. The small constants 10^{-6} and 10^{-3} are numerical safeguards that avoid zero growth and avoid a carrying capacity equal to an initial value.

The raw parameters are initialized by applying the inverse softplus map. In the implementation, this gives an initial growth rate near 3.0 on the normalized time scale, and an initial carrying capacity slightly above the largest observed response. At each training epoch the code applies the softplus transformation and evaluates the exact logistic solution for all trials and all observation times.

The fitting criterion is the masked original-scale mean squared error

$$\mathcal{L}_{\text{logistic}}(r, K) = \frac{1}{n} \sum_{(i,j) \in \Omega} \left(y_i^{[j]} - \hat{y}^{[j]}(\tau_i) \right)^2.$$

This can also be interpreted as a maximum-likelihood fit after adding a Gaussian observation-error model on the original response scale:

$$y_i^{[j]} = \hat{y}^{[j]}(\tau_i; r, K) + \varepsilon_i^{[j]}, \quad \varepsilon_i^{[j]} \stackrel{\text{ind}}{\sim} \mathcal{N}(0, \sigma^2).$$

For fixed σ^2 , maximizing this Gaussian likelihood over r and K is equivalent to minimizing the sum of squared residuals. If σ^2 is also unknown, profiling it out gives

$$\hat{\sigma}^2(r, K) = \frac{1}{n} \sum_{(i,j) \in \Omega} \left(y_i^{[j]} - \hat{y}^{[j]}(\tau_i; r, K) \right)^2,$$

and the maximizing values of r and K are again the nonlinear least-squares solution. Therefore the implemented classical fit is a profiled Gaussian maximum-likelihood estimate under an independent additive measurement-error assumption. It is not a transition likelihood for a stochastic logistic process.

The code backpropagates through the closed-form expression above and applies an Adam update [5] to a and b . No numerical ODE solver is used for this classical baseline. This model was trained for 3000 Adam epochs with learning rate 5×10^{-2} .

The inverse-problem procedure for this classical logistic model is therefore:

1. Read the observed trial matrix and keep only entries in Ω .
2. Fix the initial condition of each trial at its observed first value $y_0^{[j]}$; these initial values are not optimized.
3. Choose unconstrained optimization variables a and b , and transform them into admissible physical parameters $r(a) > 0$ and $K(b) > K_{\min}$ with the softplus maps above.
4. For the current values of a and b , compute the exact logistic curve $\hat{y}^{[j]}(\tau_i; r, K)$ for every trial and observation time.
5. Evaluate the masked residual objective $\mathcal{L}_{\text{logistic}}(r, K)$, using only observed entries.
6. Use automatic differentiation through the closed-form solution to compute $\partial \mathcal{L}_{\text{logistic}} / \partial a$ and $\partial \mathcal{L}_{\text{logistic}} / \partial b$.
7. Apply an Adam step to a and b , then repeat until the fixed epoch budget is reached.
8. Report $\hat{r} = r(\hat{a})$ and $\hat{K} = K(\hat{b})$, and evaluate fitted curves and residual metrics on the original data scale.

In this baseline the inverse problem is low-dimensional: only two unknown physical parameters are estimated. The use of gradient descent is an implementation choice; mathematically, the estimator is the nonlinear least-squares solution for the two-parameter logistic curve under fixed trial-specific initial conditions. The fitted values were

$$\hat{r} = 7.705769, \quad \hat{K} = 47.860203.$$

Since $\tau = t/168$ for these data, the growth rate in the original time scale is approximately $\hat{r}/168 = 0.045868$ per original time unit.

2.2 Neural ODE

The deterministic neural model is the non-autonomous Neural ODE [2]

$$\frac{dz}{d\tau} = f_{\theta}(\tau, z), \quad z(0) = z_0.$$

The fitted solution is a single population curve, not one curve per trial. The three trials are treated as replicate observations of the same underlying trajectory. On the original response scale the initial condition is

$$y_0 = y(t_0) = \frac{1}{J} \sum_{j=1}^J y_0^{[j]},$$

the mean of the first observed values across trials. The corresponding normalized initial condition is

$$z_0 = \frac{y_0 - \bar{y}}{s_y}.$$

After solving the normalized ODE, the fitted value on the original scale is

$$\hat{y}_i = \hat{y}(t_i) = \bar{y} + s_y \hat{z}(\tau_i; \theta).$$

The same value $\hat{y}_i$ is compared with every trial observation $y_i^{[j]}$ at time t_i .

The right-hand side f_θ is a fully connected neural network with input $(\tau, z) \in \mathbb{R}^2$, two hidden layers, 32 hidden units per layer, and hyperbolic tangent activation:

$$h_1 = \tanh(W_1 x + b_1), \quad h_\ell = \tanh(W_\ell h_{\ell-1} + b_\ell), \quad f_\theta(x) = W_o h_2 + b_o,$$

where $x = (\tau, z)^T$. The network has 1185 trainable weights and biases. This count comes from the three affine layers:

$$\begin{aligned} \text{input to hidden 1: } & 2 \cdot 32 + 32 = 96, \\ \text{hidden 1 to hidden 2: } & 32 \cdot 32 + 32 = 1056, \\ \text{hidden 2 to output: } & 32 \cdot 1 + 1 = 33. \end{aligned}$$

Thus $96 + 1056 + 33 = 1185$. The architecture was chosen as a compact smooth function approximator for a scalar time-dependent vector field. The input has only two coordinates, time and state, and the data set has only 45 scalar observations, so a very large network would make the comparison dominated by overparameterization. Two hidden layers with 32 units give substantially more flexibility than the two-parameter logistic curve while remaining small enough to train stably with weight decay. The tanh nonlinearity gives a smooth right-hand side, which is useful for ODE/SDE integration, and sigmoidal neural networks are classical universal approximators [3]. The same architecture is used for the Neural ODE and for each Neural SDE coefficient so that differences between the deterministic and stochastic neural models are not caused by different network sizes.

The ODE was solved at the observed normalized times using RK4 with step size 0.01. The empirical loss was

$$\mathcal{L}_{\text{ODE}}(\theta) = \frac{1}{n} \sum_{(i,j) \in \Omega} \left(z_i^{[j]} - \hat{z}_i(\theta) \right)^2.$$

Equivalently, in original response units this loss fits one curve $\hat{y}_i$ to all three replicate measurements at the same time. The minimizer is therefore encouraged to pass through the center of the replicate cloud, while the ODE structure enforces a smooth dynamical trajectory. Training used Adam for 1000 epochs with learning rate 10^{-2} and weight decay 10^{-4} [5].

2.3 Neural SDE

The stochastic neural model is an Ito diffusion with neural drift and diffusion coefficients, following the Neural SDE modeling idea [9, 4]:

$$dZ_\tau = \mu_\theta(\tau, Z_\tau) d\tau + \sigma_\phi(\tau, Z_\tau) dW_\tau.$$

Both the drift μ_θ and the raw diffusion network use the same architecture as the Neural ODE: two hidden layers, 32 units per hidden layer, and tanh activation. The diffusion is constrained to be positive by

$$\sigma_\phi(\tau, z) = \text{softplus}(g_\phi(\tau, z)) + \sigma_{\min}, \quad \sigma_{\min} = 10^{-3}.$$

The total trainable parameter count is 2370, one copy of the 1185-parameter network for the drift and one for the diffusion.

The SDE was trained with an Euler-Maruyama Gaussian transition likelihood [6]. For successive observations separated by $\Delta\tau_i = \tau_{i+1} - \tau_i$, the approximation is

$$Z_{i+1}^{[j]} \mid Z_i^{[j]} \sim \mathcal{N}\left(Z_i^{[j]} + \mu_\theta(\tau_i, Z_i^{[j]})\Delta\tau_i, \sigma_\phi(\tau_i, Z_i^{[j]})^2\Delta\tau_i\right).$$

With the variance floor used in the implementation, define

$$m_i^{[j]} = z_i^{[j]} + \mu_\theta(\tau_i, z_i^{[j]})\Delta\tau_i, \quad v_i^{[j]} = \sigma_\phi(\tau_i, z_i^{[j]})^2\Delta\tau_i + 10^{-6}.$$

The minimized negative log-likelihood is

$$\mathcal{L}_{\text{SDE}}(\theta, \phi) = \frac{1}{|\Omega_{\text{tr}}|} \sum_{(i,j) \in \Omega_{\text{tr}}} \frac{1}{2} \left[\log(2\pi v_i^{[j]}) + \frac{(z_{i+1}^{[j]} - m_i^{[j]})^2}{v_i^{[j]}} \right],$$

where Ω_{tr} is the set of adjacent observed transitions. Training used Adam for 1000 epochs with learning rate 5×10^{-3} and weight decay 10^{-4} [5]. After training, 1000 Monte Carlo paths were simulated to estimate predictive means and central 90 percent intervals.

2.4 Logistic PINN inverse models

The two new models solve the inverse problem for the logistic growth law with a physics-informed neural network (PINN) [7]. Unlike the classical logistic baseline, the PINN does not evaluate the closed-form logistic solution during training. Instead, it learns a neural trajectory $u_\eta(\tau)$ and estimates the physical parameters by penalizing violations of the differential equation.

The unknowns in the deterministic PINN inverse problem are

$$\eta, \quad r, \quad K,$$

where η denotes all neural-network weights and biases. The unknowns in the stochastic PINN inverse problem are

$$\eta, \quad r, \quad K, \quad \sigma.$$

The role of the neural trajectory u_η is to provide a differentiable surrogate for the unobserved continuous-time biomass curve. The physical parameters are then identified by forcing this differentiable surrogate to match the measurements and to satisfy the logistic differential equation at collocation points between measurements. This is the main distinction from the closed-form classical logistic fit: the PINN estimates a curve and the parameters jointly.

2.4.1 Shared PINN trajectory architecture

Both logistic PINNs use a scalar neural trajectory

$$u_\eta : [0, 1] \rightarrow \mathbb{R}_+, \quad \hat{y}(t) = u_\eta(\tau(t)).$$

The network input is the normalized time τ . The hidden architecture has three fully connected hidden layers, 32 units in each hidden layer, and hyperbolic tangent activation:

$$h_1 = \tanh(W_1\tau + b_1), \quad h_\ell = \tanh(W_\ell h_{\ell-1} + b_\ell), \quad \ell = 2, 3,$$

followed by one scalar output layer. Positivity of the fitted biomass trajectory is enforced by

$$u_\eta(\tau) = \text{softplus}(W_o h_3 + b_o) + 10^{-6}.$$

The trajectory network has 2209 trainable weights and biases:

$$\begin{aligned} \text{input to hidden 1:} & \quad 1 \cdot 32 + 32 = 64, \\ \text{hidden 1 to hidden 2:} & \quad 32 \cdot 32 + 32 = 1056, \\ \text{hidden 2 to hidden 3:} & \quad 32 \cdot 32 + 32 = 1056, \\ \text{hidden 3 to output:} & \quad 32 \cdot 1 + 1 = 33. \end{aligned}$$

Thus the deterministic logistic PINN has $2209 + 2 = 2211$ fitted parameters after adding the raw variables for r and K , while the stochastic logistic PINN has $2209 + 3 = 2212$ fitted parameters after adding the raw variable for σ .

The carrying capacity is constrained to remain above all observed responses, not only above the first row. Define

$$K_{\min} = \max_{(i,j) \in \Omega} y_i^{[j]} + 10^{-3}.$$

The deterministic PINN optimizes unconstrained variables $a, b \in \mathbb{R}$ and maps them to physical parameters by

$$r = \text{softplus}(a) + 10^{-8}, \quad K = K_{\min} + \text{softplus}(b) + 10^{-8}.$$

The stochastic PINN adds a third unconstrained variable c and uses

$$\sigma = \text{softplus}(c) + 10^{-8}.$$

Here r is on the original time scale, because the physics residual includes the time-scale factor $t_{\max} - t_{\min} = 168$.

The PINN initialization is chosen to start from a stable positive trajectory. The final network layer is initialized so that $u_\eta(\tau)$ is approximately constant at the mean observed initial level,

$$\bar{y}_0 = \frac{1}{J} \sum_{j=1}^J y_0^{[j]}.$$

The initial carrying capacity is set slightly above the largest observed response, and the initial growth rate is estimated from the first and last observed means using the analytic logistic relationship. If that estimate is not admissible, a small positive default value is used. The stochastic fit starts from a positive multiplicative diffusion value, $\sigma = 0.02$, before gradient-based optimization.

2.4.2 Deterministic logistic PINN

The deterministic inverse problem is based on the original-time logistic ODE

$$\frac{dX}{dt} = rX \left(1 - \frac{X}{K}\right).$$

Since the PINN is differentiated with respect to normalized time τ , the residual uses

$$\frac{du_\eta}{d\tau} = (t_{\max} - t_{\min}) r u_\eta(\tau) \left(1 - \frac{u_\eta(\tau)}{K}\right).$$

Let $\{\xi_m\}_{m=1}^M$ be $M = 128$ uniformly spaced collocation points in $[0, 1]$. The data, initial-condition, and physics losses are

$$\begin{aligned}\mathcal{L}_{\text{data}}(\eta) &= \frac{1}{n} \sum_{(i,j) \in \Omega} \left(\frac{u_\eta(\tau_i) - y_i^{[j]}}{s_y} \right)^2, \\ \mathcal{L}_{\text{init}}(\eta) &= \frac{1}{J} \sum_{j=1}^J \left(\frac{u_\eta(0) - y_0^{[j]}}{s_y} \right)^2,\end{aligned}$$

and

$$\mathcal{L}_{\text{phys}}(\eta, r, K) = \frac{1}{M} \sum_{m=1}^M \left[\frac{\frac{du_\eta}{d\tau}(\xi_m) - (t_{\max} - t_{\min}) r u_\eta(\xi_m) \left(1 - \frac{u_\eta(\xi_m)}{K}\right)}{s_y} \right]^2.$$

The deterministic PINN objective is

$$\mathcal{L}_{\text{detPINN}} = \mathcal{L}_{\text{data}} + \lambda_0 \mathcal{L}_{\text{init}} + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}},$$

with $\lambda_0 = 1$ and $\lambda_{\text{phys}} = 1$ in the reported run. This fit uses one population trajectory against all three replicate trials. It is therefore more flexible than the classical two-parameter logistic regression, because the neural trajectory can absorb deviations from the exact analytic curve, but it is still regularized toward the logistic differential equation. Training used Adam for 3000 epochs with learning rate 5×10^{-3} .

The deterministic PINN inverse problem is solved by the following optimization loop:

1. Build the observation tensor, mask missing entries, compute s_y , and map observation times to $\tau_i \in [0, 1]$.
2. Create the collocation grid $\xi_1, \dots, \xi_M$ on $[0, 1]$. These points are not additional data; they are locations where the logistic differential equation is enforced.
3. Initialize the trajectory network u_η and raw physical variables a, b . The reported parameters at every epoch are obtained from $r = \text{softplus}(a) + 10^{-8}$ and $K = K_{\min} + \text{softplus}(b) + 10^{-8}$.
4. Evaluate the population trajectory at the observation times to compute $\mathcal{L}_{\text{data}}$ and $\mathcal{L}_{\text{init}}$.
5. Evaluate $u_\eta(\xi_m)$ at the collocation points and use automatic differentiation to compute $du_\eta/d\tau$ at those same points.
6. Form the residual $du_\eta/d\tau - (t_{\max} - t_{\min}) r u_\eta(1 - u_\eta/K)$ and compute $\mathcal{L}_{\text{phys}}$.
7. Add the weighted terms to obtain $\mathcal{L}_{\text{detPINN}}$, backpropagate through both the network and the softplus-transformed parameters, and update (η, a, b) with Adam.
8. After training, transform the final raw variables to $\hat{r}$ and $\hat{K}$, and evaluate $u_{\hat{\eta}}(\tau)$ on the observed and dense plotting grids.

The gradients with respect to r and K are not computed by finite differences. They are obtained by automatic differentiation through the loss terms. In particular, the physics loss connects r and K directly to the neural trajectory derivative, so the fitted parameters are those that make the learned curve both close to the observations and dynamically consistent with logistic growth.

2.4.3 Stochastic logistic PINN/SDE

The stochastic inverse problem augments the same logistic drift with multiplicative process noise:

$$dX_t = rX_t \left(1 - \frac{X_t}{K}\right) dt + \sigma X_t dW_t.$$

The stochastic PINN retains the same neural trajectory, data loss, initial loss, and physics residual used by the deterministic PINN. The additional identification of σ comes from an Euler-Maruyama transition likelihood on adjacent observed pairs. For $\Delta t_i = t_{i+1} - t_i$,

$$Y_{i+1}^{[j]} \mid Y_i^{[j]} \approx \mathcal{N}(m_i^{[j]}, v_i^{[j]}),$$

where

$$m_i^{[j]} = y_i^{[j]} + ry_i^{[j]} \left(1 - \frac{y_i^{[j]}}{K}\right) \Delta t_i, \quad v_i^{[j]} = \left(\sigma y_i^{[j]}\right)^2 \Delta t_i + 10^{-6}.$$

Let Ω_{tr} be the set of adjacent observed transitions. The transition negative log-likelihood is

$$\mathcal{L}_{\text{tr}}(r, K, \sigma) = \frac{1}{|\Omega_{\text{tr}}|} \sum_{(i,j) \in \Omega_{\text{tr}}} \frac{1}{2} \left[\log(2\pi v_i^{[j]}) + \frac{(y_{i+1}^{[j]} - m_i^{[j]})^2}{v_i^{[j]}} \right].$$

Operationally, this loss is computed one observed step at a time. For each complete adjacent pair, the current value $y_i^{[j]}$ and time increment Δt_i determine the Euler-Maruyama mean $m_i^{[j]}$ and variance $v_i^{[j]}$. The difference $y_{i+1}^{[j]} - m_i^{[j]}$ is the one-step innovation around the logistic drift. The Gaussian negative log-likelihood then combines two penalties: a scaled squared innovation $(y_{i+1}^{[j]} - m_i^{[j]})^2 / v_i^{[j]}$, which penalizes transitions that are far from the predicted mean, and a $\log(2\pi v_i^{[j]})$ term, which penalizes unnecessarily large transition variance. The reported transition NLL is the average of these per-transition contributions over Ω_{tr} . Thus a smaller value means that the observed kefir growth increments are more plausible under the stochastic logistic dynamics. The parameter σ is identified through this term because it controls $v_i^{[j]}$: if σ is too small, ordinary short-time fluctuations are assigned very low probability, whereas if σ is too large, the log-variance part of the likelihood penalizes the excess uncertainty.

The total stochastic objective is

$$\mathcal{L}_{\text{stoPINN}} = \mathcal{L}_{\text{data}} + \lambda_0 \mathcal{L}_{\text{init}} + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}} + \lambda_{\text{tr}} \mathcal{L}_{\text{tr}},$$

with $\lambda_0 = 1$, $\lambda_{\text{phys}} = 1$, and $\lambda_{\text{tr}} = 0.1$. The transition likelihood is evaluated on the original response scale and the original time increments, while the collocation residual is evaluated in normalized time. Training used Adam for 3000 epochs with learning rate 5×10^{-3} .

The stochastic PINN/SDE estimation procedure extends the deterministic loop as follows:

1. Construct the same observation losses and collocation physics loss used by the deterministic PINN.
2. Build a transition data set from adjacent observations: $(y_i^{[j]}, y_{i+1}^{[j]}, \Delta t_i)$ is included only when both endpoints are observed.
3. Optimize three raw physical variables a, b, c , transformed to $r > 0$, $K > K_{\text{min}}$, and $\sigma > 0$ with softplus maps.

4. For every adjacent transition, compute the Euler-Maruyama conditional mean $m_i^{[j]}$ and variance $v_i^{[j]}$ implied by the current r , K , and σ .
5. Compute the Gaussian transition negative log-likelihood $\mathcal{L}_{\text{tr}}$. This term estimates the diffusion intensity because σ controls the transition variance.
6. Add the weighted transition likelihood to the PINN data, initial, and physics losses, then backpropagate through the full objective to update (η, a, b, c) .
7. After training, report $\hat{r} = r(\hat{a})$, $\hat{K} = K(\hat{b})$, and $\hat{\sigma} = \sigma(\hat{c})$.

The stochastic fit uses two complementary sources of information. The PINN terms estimate a smooth population drift trajectory and keep that trajectory close to logistic growth. The transition likelihood estimates how much short-time variability remains around the logistic drift. Consequently, σ is identified from the observed increments, not from the pointwise distance between the PINN curve and the data.

After training, the stochastic logistic SDE was simulated with Euler-Maruyama. Each trial-specific simulation starts from its observed initial value $y_0^{[j]}$. The reported stochastic mean and central 90 percent predictive interval use 5000 Monte Carlo paths. In the numerical table, “Stochastic Logistic PINN Drift” refers to the fitted neural trajectory u_η , whereas “Stochastic Logistic SDE Mean” refers to the Monte Carlo mean of the fitted SDE.

3 Comparison criteria

All model comparison metrics were computed on the original response scale using the fitted mean trajectory for each model. For models that produce one curve per trial, write this fitted value as $\hat{y}_i^{[j]}$. For the single-curve Neural ODE, $\hat{y}_i^{[j]} = \hat{y}_i$ for every trial j . Let

$$\text{SSE} = \sum_{(i,j) \in \Omega} (y_i^{[j]} - \hat{y}_i^{[j]})^2, \quad \text{SST} = \sum_{(i,j) \in \Omega} (y_i^{[j]} - \bar{y})^2.$$

The root mean squared error is

$$\text{RMSE} = \sqrt{\frac{\text{SSE}}{n}},$$

and the coefficient of determination is

$$R^2 = 1 - \frac{\text{SSE}}{\text{SST}}.$$

The numerical table below reports in-sample residual diagnostics: the models are trained on all observed entries in Ω , and the same entries are used to compute RMSE and R^2 . This is appropriate for checking interpolation quality, but it is not an out-of-sample generalization error. To estimate a test MSE for this time-series problem, the split should be made by time rather than by randomly sampling individual observations. Let

$$I_{\text{train}} \subset \{0, \dots, T-1\}, \quad I_{\text{test}} \subset \{0, \dots, T-1\}, \quad I_{\text{train}} \cap I_{\text{test}} = \emptyset,$$

and define

$$\Omega_{\text{train}} = \{(i, j) \in \Omega : i \in I_{\text{train}}\}, \quad \Omega_{\text{test}} = \{(i, j) \in \Omega : i \in I_{\text{test}}\}.$$

The Neural ODE is then trained by replacing Ω with Ω_{train} in $\mathcal{L}_{\text{ODE}}(\theta)$, while the test MSE is computed as

$$\text{MSE}_{\text{test}} = \frac{1}{|\Omega_{\text{test}}|} \sum_{(i,j) \in \Omega_{\text{test}}} \left(y_i^{[j]} - \hat{y}_i \right)^2.$$

For these data, a natural forecasting split is to train on early times and test on the last few time points, for example training through $t = 132$ and testing at $t = 144, 156, 168$. With only 15 time points, an alternative is a rolling leave-last-time-out protocol, such as training through $t = 120$ and testing at $t = 132$, then training through $t = 132$ and testing at $t = 144$, and so on.

For AIC and BIC [1, 8], residuals were modeled as independent Gaussian errors with common variance σ^2 . The maximum likelihood variance estimate is

$$\hat{\sigma}^2 = \frac{\text{SSE}}{n},$$

which gives the maximized log-likelihood

$$\log \hat{L} = -\frac{n}{2} [\log(2\pi) + 1 + \log(\hat{\sigma}^2)].$$

The criteria are

$$\text{AIC} = 2k - 2 \log \hat{L}, \quad \text{BIC} = k \log(n) - 2 \log \hat{L}.$$

Here $k = p + 1$, where p is the number of fitted dynamical parameters and the additional parameter is the residual variance. Thus $k = 3$ for the classical logistic model, $k = 1186$ for the Neural ODE, $k = 2371$ for the Neural SDE, $k = 2212$ for the deterministic logistic PINN, and $k = 2213$ for the stochastic logistic PINN/SDE. Initial conditions are not counted as fitted parameters because they are taken directly from the first row of the data; for the single-curve Neural ODE this means using $y_0 = J^{-1} \sum_{j=1}^J y_0^{[j]}$.

The 90 percent interval coverage is computed for models that produce a predictive distribution rather than only a fitted mean curve: the Neural SDE and the stochastic logistic SDE. Let B be the number of simulated SDE paths ($B = 1000$ for the Neural SDE comparison and $B = 5000$ for the stochastic logistic SDE) and let $Y_i^{(b)[j]}$ denote the simulated original-scale value for path b , time i , and trial j . For interval level $\gamma = 0.90$, set $\alpha = (1 - \gamma)/2 = 0.05$ and define the pointwise empirical interval

$$L_i^{[j]} = Q_\alpha \left(\{Y_i^{(b)[j]}\}_{b=1}^B \right), \quad U_i^{[j]} = Q_{1-\alpha} \left(\{Y_i^{(b)[j]}\}_{b=1}^B \right),$$

where Q_q is the empirical q -quantile. The reported empirical coverage is

$$\hat{C}_{0.90} = \frac{1}{n} \sum_{(i,j) \in \Omega} \mathbf{1}\{L_i^{[j]} \leq y_i^{[j]} \leq U_i^{[j]}\}.$$

Thus an observed data point is counted as covered if it lies between the 5 percent and 95 percent Monte Carlo quantiles for the same time and trial. For the Neural SDE, the reported value 0.9333 means that 42 of the 45 observed responses fall inside their corresponding central 90 percent intervals. For the stochastic logistic SDE, the reported value 0.8222 means that 37 of the 45 observed responses fall inside their corresponding central 90 percent intervals. The mean interval width is the average of $U_i^{[j]} - L_i^{[j]}$ over the same observed entries. These are in-sample empirical calibration diagnostics, not out-of-sample guarantees.

Table 1: Model comparison on the original response scale. Lower RMSE, AIC, and BIC are better; larger R^2 is better. The PINN AIC/BIC values use the same Gaussian residual criterion as the other fitted mean curves.

Model	RMSE	R^2	AIC	BIC	p	k
Classical logistic ODE	1.4291	0.9826	165.84	171.26	2	3
Neural ODE	1.0281	0.9910	2502.20	4644.90	1185	1186
Neural SDE	2.0312	0.9649	4933.48	9217.08	2370	2371
Deterministic logistic PINN	2.8167	0.9326	4644.90	8641.24	2211	2212
Stochastic logistic PINN drift	2.8469	0.9311	4647.87	8646.01	2212	2213
Stochastic logistic SDE mean	3.2324	0.9112	4659.30	8657.44	2212	2213

Table 2: Additional stochastic-model diagnostics.

Model	90 percent interval coverage	Mean interval width
Neural SDE	0.9333	7.3887
Stochastic logistic SDE	0.8222	9.4517

Table 3: Estimated logistic PINN inverse-problem parameters. The growth rate r is on the original time scale.

Model	r	K	σ
Deterministic logistic PINN	0.035062	52.3350	–
Stochastic logistic PINN/SDE	0.036806	52.2809	0.017245

4 Numerical results

The Neural ODE has the best in-sample interpolation accuracy, with the smallest RMSE and largest R^2 . The classical logistic ODE is less flexible, but it still explains approximately 98.3 percent of the observed variance. Because AIC and BIC penalize the number of fitted parameters, the two-parameter logistic model is strongly preferred by these information criteria.

The deterministic logistic PINN and the stochastic logistic PINN drift recover similar logistic parameters, with $r \approx 0.035$ – 0.037 on the original time scale and $K \approx 52.3$. Their RMSE values are larger than those of the classical logistic ODE and the Neural ODE because the PINN objective balances data fit, initial-condition fit, and a collocation residual rather than minimizing only the closed-form logistic least-squares objective. The stochastic logistic SDE mean is less accurate as a fitted mean curve than the deterministic PINN trajectory, but it provides a process-noise estimate and a central 90 percent predictive band.

The AIC and BIC values for the neural models should be interpreted as parameter-penalized residual criteria for the fitted mean curves. They are not the same as comparing the SDE transition likelihood directly to an exact ODE likelihood. For the PINNs, these criteria also penalize every neural-network weight even though the scientific inverse problem has only two or three physical parameters. Their practical role here is to quantify the tradeoff between in-sample residual accuracy and model complexity, not to declare a definitive mechanistic winner.

All comparison figures below were regenerated with the shared plotting style in `src/kefir_models/plot_style.py`. Observed trial markers, fitted mean curves, predictive bands, legend placement, and the fixed tight canvas are therefore consistent across the Neural ODE/SDE, logistic PINN, and combined all-model figures. The step-by-step PNG sequences generated by the plotting scripts use the same canvas dimensions, so adding a legend entry does not shift the plotted data region.

5 Reproducibility

The original classical, Neural ODE, and Neural SDE comparison was generated by the SDE comparison command-line tool implemented in `src/kefir_models/sde_compare.py`. The deterministic and stochastic logistic PINN comparison was generated by the logistic PINN command-line tool implemented in `src/kefir_models/logistic_pinn_compare.py`. The data file used in this directory is named `data/raw/waterKefirTrialsReferece.csv`. The following command reproduces the original Neural ODE/SDE comparison from the project root:

```
kefir-sde-compare \
  --data-path data/raw/waterKefirTrialsReferece.csv \
  --output-dir outputs/neural_sde_comparison_outputs \
  --classical-epochs 3000 \
  --classical-learning-rate 5e-2 \
  --ode-epochs 1000 \
  --ode-learning-rate 1e-2 \
  --sde-epochs 1000 \
  --sde-learning-rate 5e-3 \
  --hidden-units 32 \
  --hidden-layers 2 \
  --weight-decay 1e-4 \
  --min-diffusion 1e-3 \
```

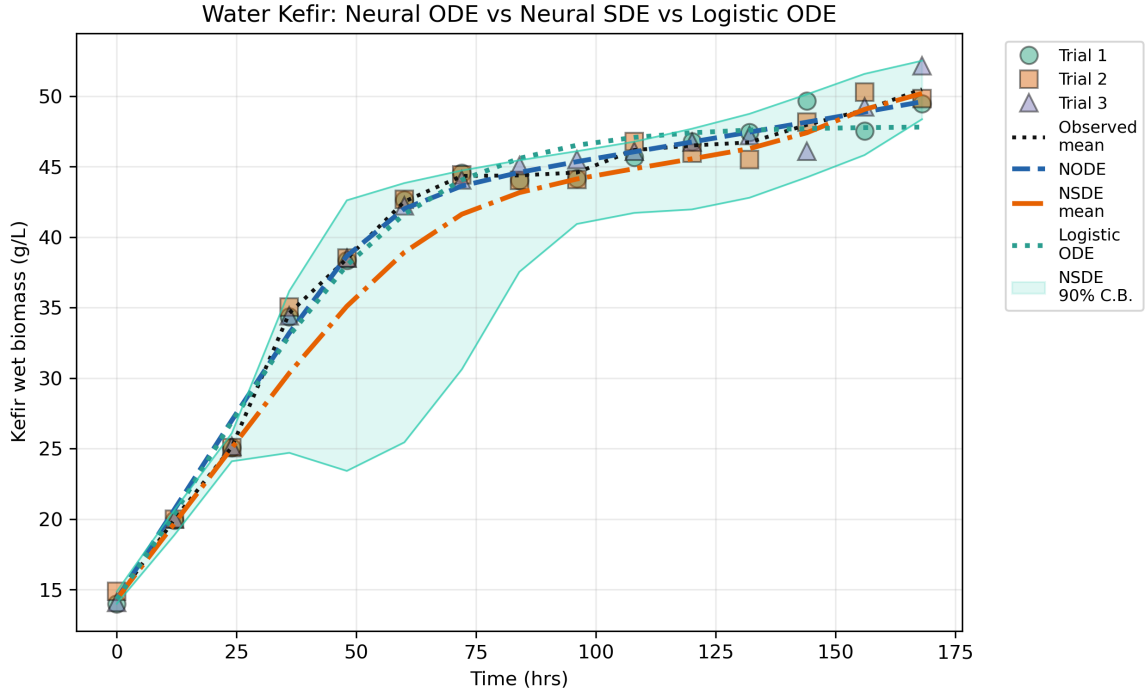


Figure 1: Observed trial data, the single Neural ODE population curve, fitted mean trajectories from the comparison models, and the Neural SDE central 90 percent predictive interval. The Neural ODE is fitted as one curve against all three trial realizations, while the logistic ODE captures the saturating growth pattern with a much smaller parameter count. The figure uses the shared fixed-canvas comparison layout.

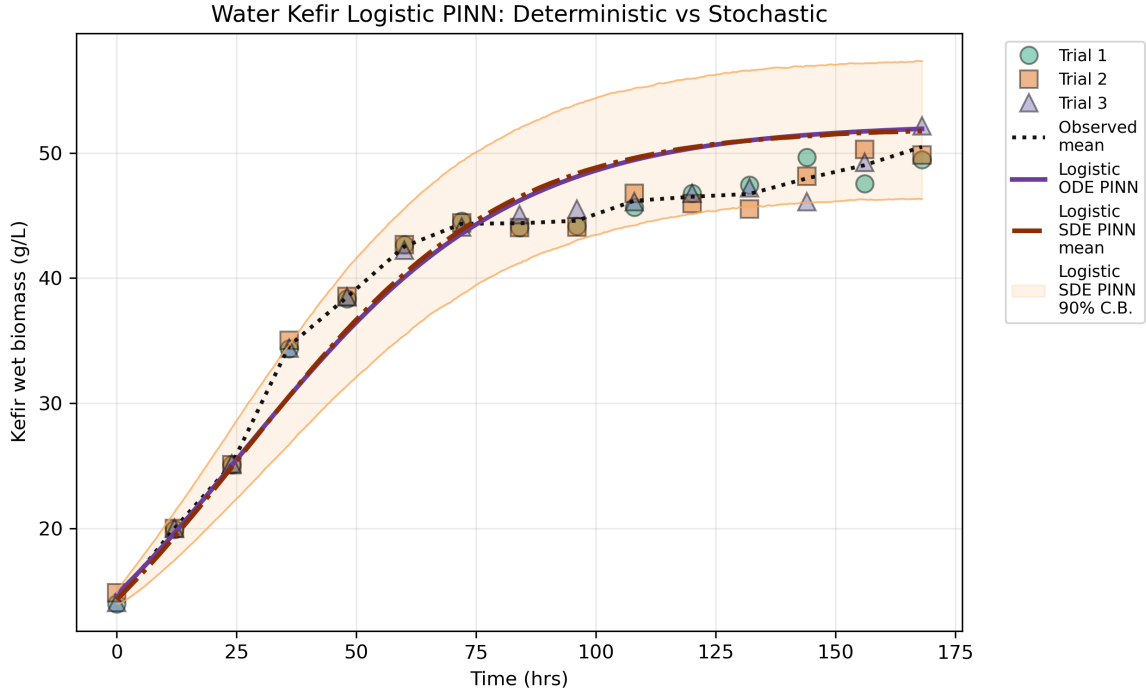


Figure 2: Observed trial data, deterministic logistic PINN population trajectory, stochastic logistic SDE predictive mean, and central 90 percent predictive interval. The deterministic and stochastic PINNs use the same three-hidden-layer trajectory architecture and differ by the stochastic transition-likelihood term used to identify σ . The line colors and styles match the all-model comparison.

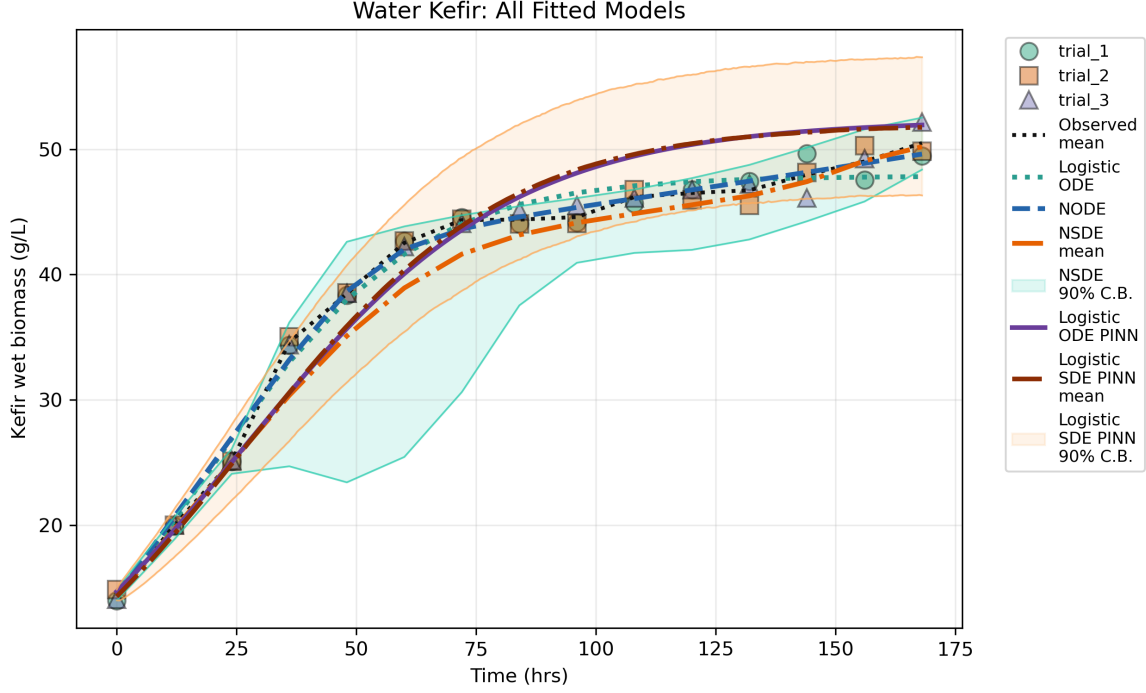


Figure 3: Combined comparison of the classical logistic ODE, Neural ODE, Neural SDE, deterministic logistic PINN, and stochastic logistic PINN/SDE outputs. The plot overlays the two previously separate comparison sequences on a common original response scale so that fitted mean trajectories and predictive bands can be visually compared without a layout change between sequence frames.

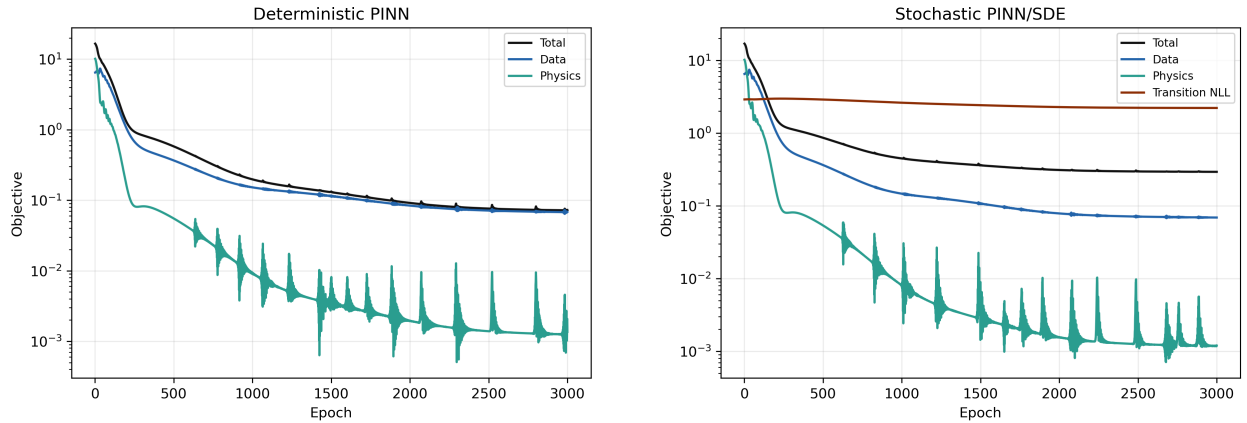


Figure 4: Training objectives for the deterministic and stochastic logistic PINNs. The deterministic objective combines normalized data, initial-condition, and physics residual terms. The stochastic objective adds the weighted Euler-Maruyama transition negative log-likelihood.

Training Objective Histories by Scale

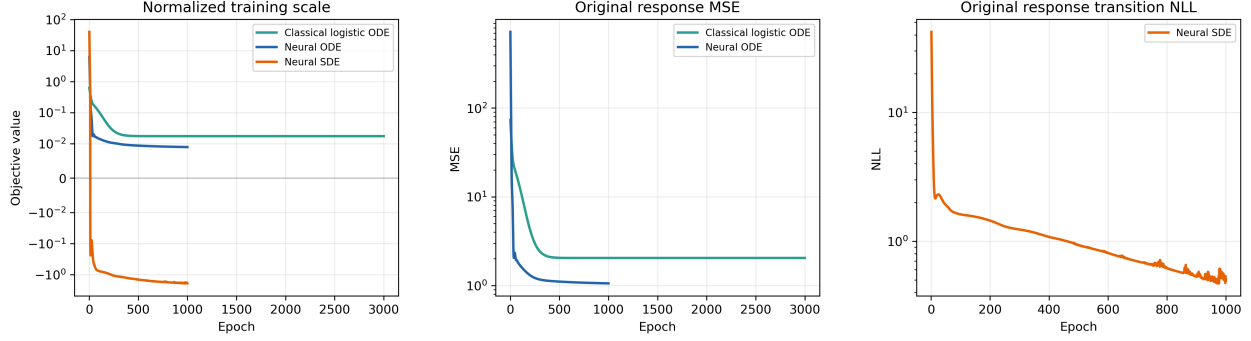


Figure 5: Training objective histories from the saved experiment outputs, re-expressed by scale. The first panel shows the objectives on normalized training scales, the second panel compares the classical and Neural ODE MSE histories on the original response scale, and the third panel shows the Neural SDE transition negative log-likelihood on the original response scale. These histories are optimization diagnostics, not direct model-comparison metrics, because MSE and transition likelihood remain different objective types.

```
--ode-method rk4 \
--sde-paths 1000 \
--interval-level 0.90 \
--seed 7 \
--device auto \
--plot
```

The following command reproduces the logistic PINN inverse-problem outputs:

```
kefir-logistic-pinn-compare \
--config logistic_pinn_config_reference.json
```

The following plotting-only command builds the third step-by-step sequence that overlays the classical logistic ODE, Neural ODE, Neural SDE, deterministic logistic PINN, and stochastic logistic PINN/SDE outputs:

```
kefir-plot-all-models \
--neural-comparison-csv \
    outputs/neural_sde_comparison_outputs/water_kefir_neural_dynamics_comparison.csv \
--pinn-comparison-csv \
    logistic_pinn_outputs/water_kefir_logistic_pinn_comparison.csv \
--pinn-dense-csv \
    logistic_pinn_outputs/water_kefir_logistic_pinn_dense.csv \
--output-dir outputs/all_model_comparison_outputs \
--interval-level 0.90
```

The updated PNGs in this report use the shared style and fixed-canvas save helpers in `src/kefir_models/plot_style.py`. The comparison figures are written with the same 2594×1590 pixel canvas, while the wider loss and objective-history diagnostics are saved with tight bounds.

Equivalently, the expanded logistic PINN configuration is:

```

kefir-logistic-pinn-compare \
  --data-path data/raw/waterKefirTrialsReferece.csv \
  --output-dir logistic_pinn_outputs \
  --deterministic-epochs 3000 \
  --stochastic-epochs 3000 \
  --learning-rate 5e-3 \
  --hidden-units 32 \
  --hidden-layers 3 \
  --collocation-points 128 \
  --physics-weight 1.0 \
  --initial-weight 1.0 \
  --transition-weight 0.1 \
  --initial-diffusion 0.02 \
  --sde-paths 5000 \
  --interval-level 0.90 \
  --plot-points 300 \
  --seed 7 \
  --device auto \
  --plot

```

The main output files used in this report are:

- outputs/neural_sde_comparison_outputs/model_comparison_metrics.csv,
- outputs/neural_sde_comparison_outputs/classical_logistic_parameters.csv,
- outputs/neural_sde_comparison_outputs/water_kefir_neural_dynamics_comparison.csv,
- outputs/neural_sde_comparison_outputs/water_kefir_neural_dynamics_comparison.png,
- outputs/neural_sde_comparison_outputs/training_objective_scales.csv,
- outputs/neural_sde_comparison_outputs/training_objectives.png,
- outputs/all_model_comparison_outputs/water_kefir_all_models_comparison.png,
- logistic_pinn_outputs/water_kefir_logistic_pinn_metrics.csv,
- logistic_pinn_outputs/water_kefir_logistic_pinn_parameters.csv,
- logistic_pinn_outputs/water_kefir_logistic_pinn_comparison.csv,
- logistic_pinn_outputs/water_kefir_logistic_pinn_comparison.png,
- logistic_pinn_outputs/water_kefir_logistic_pinn_losses.png.

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